THE MET

MET ML EXHIBITIONS

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The Problem

- MET attendance struggling since COVID-19
- MET considers selling art pieces in order to stay afloat
- Some employees laid off due to funding issues
- Remaining employees create union efforts due to heavy workloads
- Creating exhibitions can often take years
  - The current system at the MET categorizes by department, time, type, and region for over 1.5 million works

PROPOSED SOLUTION

ML Curated Exhibitions

THE MET

DATA + PREPROCESSING

DATA ACQUISITION

- MET API

1. Metadata (objectID, title, artist, department, date, medium)
2. Image files

500 images → 1000 images for more diversity

PREPROCESSING

For recommender model:

Google Vision API **Metadata + Vision text with TF-IDF** → **CLIP embeddings (text + images)** OpenAI CLIP

- Metadata text normalization/tokenization for feature build
- CLIP embeddings for both metadata text and images in the same vector space
- Built CLIP artifacts to store and serve model state reproducibly

For user queries:

- Standard NLP preprocessing: normalized, stopwords filtered, lemmatized, and parsed for simple negation intent before encoding.

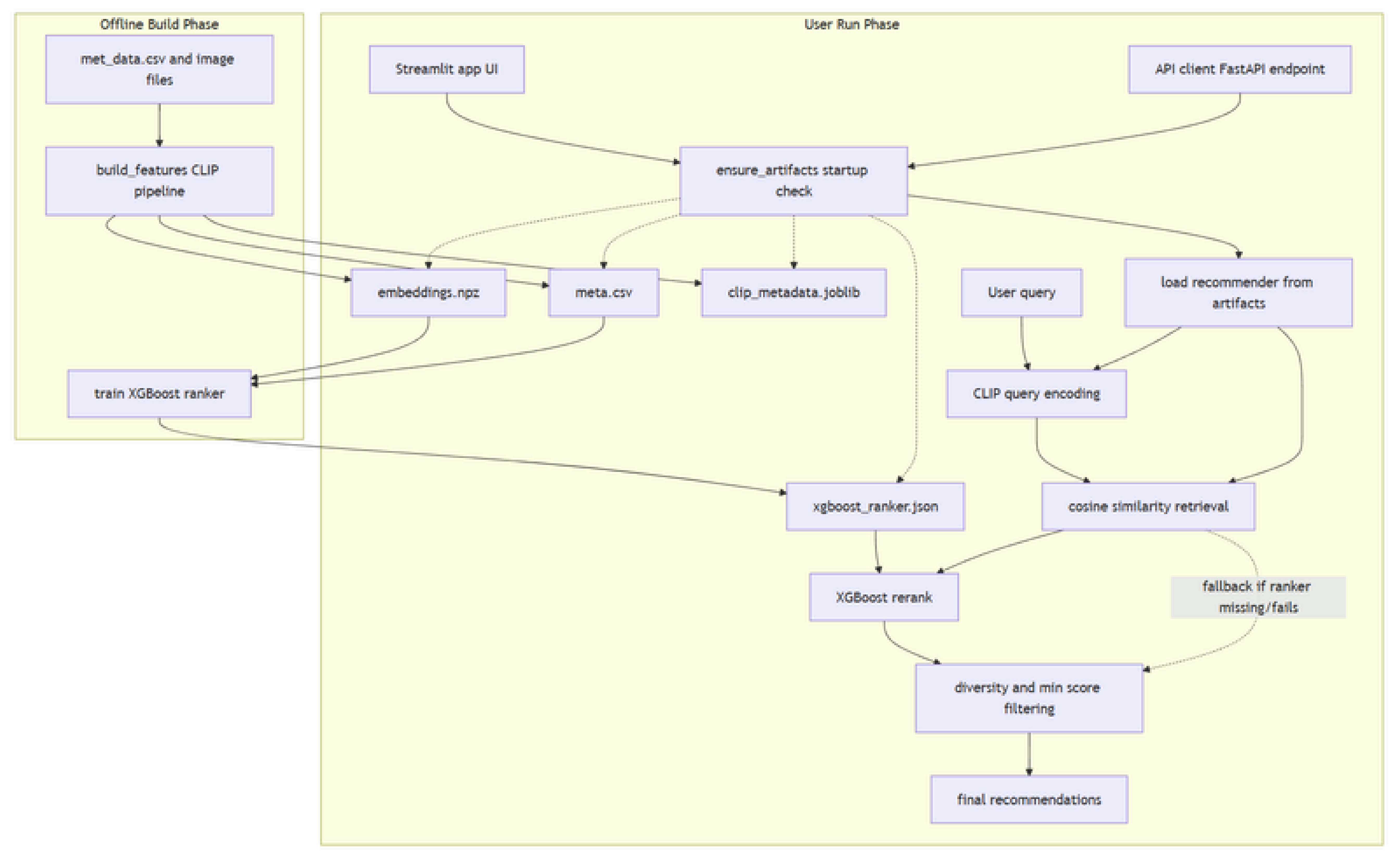
RANKING LOGIC

1. Query Intent Parsing (last slide)
   - Splits into positive tokens and negative clauses
2. Negation-Aware Scoring
   - Computes cosine similarity from positives
3. Base Embedding Scoring
   - 80% CLIP similarity + 20% metadata TF-IDF
4. XGBoost Re-Ranking Top Candidates (100)
   - Features:
     - Embedding difference
     - Cosine similarity
     - Color (RGB) difference
5. Final Output Scores
   - similarity_score → Post-negation cosine
   - raw_score → After XGBoost blend
   - score → Calibrated [0–1] display score

Main goals:

- Identify desired themes
- Combine visual AI embeddings + artwork metadata
- Consider style, time period, and visual similarity
- Balances relevance with artist & department variety

SYSTEM ARCHITECTURE




DEMO

Stop Deploy

Exhibition Setup

Theme(s) (comma-separated)

Ancient Egypt, Religious Art

Target pieces per exhibition

8

Minimum similarity

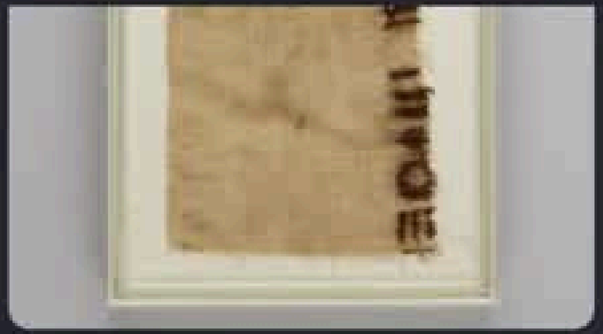
0.20

Show Selection Diagnostics

Generate

This recommender works best when your theme uses attributes represented in the collection (period, material, style, subject, color, culture, or department). If results are weak, refine your prompt with concrete

Yuny and His Wife Renenutet, Unknown | ca. 1294–1279 B.C.



Linen mark, Unknown | ca. 1051 B.C.

Amuletic plaque of Paser, the Vizier of Seti I and Ramesses II, Paser | ca. 1294–1213 B.C.

AI Selection Process & Ranking Insights

Theme Analysis: Keywords and visual patterns identified for *'Ancient Egypt'*.

Initial Candidates Found

100

Retrieved via Cosine Similarity

Final Selection

3

Refined via XGBoost Ranker

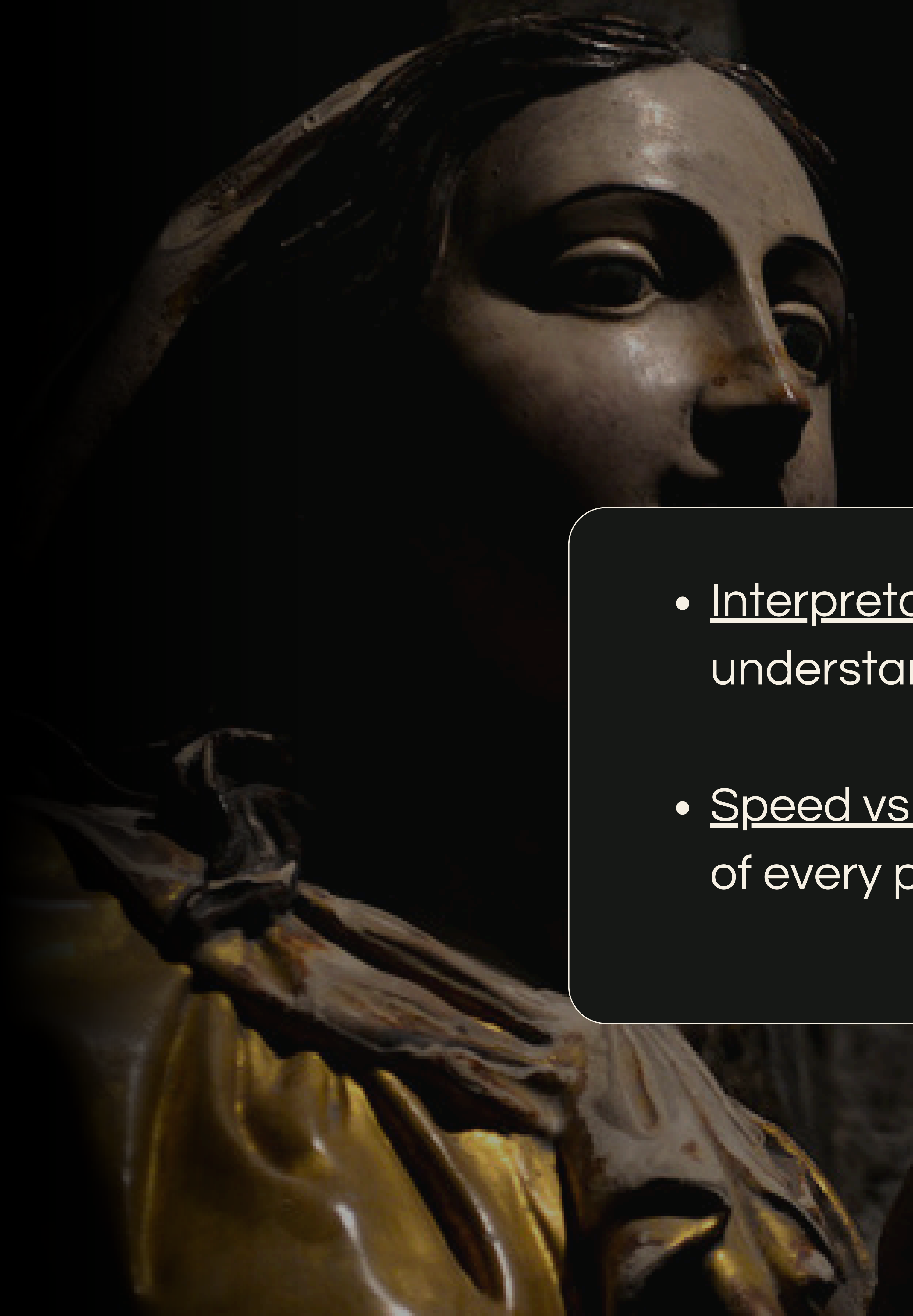
Top Ranked Selection Insights:

Artwork Title	AI Match (Raw)	Filter Bonus	Final Score
Yuny and His Wife Renenutet	0.438	+0.562	1.000
Linen mark	0.359	+0.258	0.617

EVALUATION


```
{
  "model_family": "xgboost",
  "objective": "rank:ndcg",
  "n_samples_total": 4000,
  "n_samples_train": 3200,
  "n_samples_val": 800,
  "n_queries_total": 1000,
  "n_queries_train": 800,
  "n_queries_val": 200,
  "positive_rate_train": 0.25,
  "best_iteration": -1,
  "feature_dim": 526,
```


- 4000 pairs (queries + artwork combinations)
- 1000 possible queries (generated through metadata)
- Only 25% of the artworks were relevant for each query
- Each artwork represented by a vector of 526 numbers



LIMITATIONS

- <u>Interpretability trade-off</u>: AI embeddings can make it difficult to understand why a certain decision was made
- <u>Speed vs storage trade-off</u>: Currently keeps images and artifacts of every piece, can become large when using all 1.5 million works


THANKS

GITHUB

https://github.com/romane-lg/Met_ML_Exhibitions

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